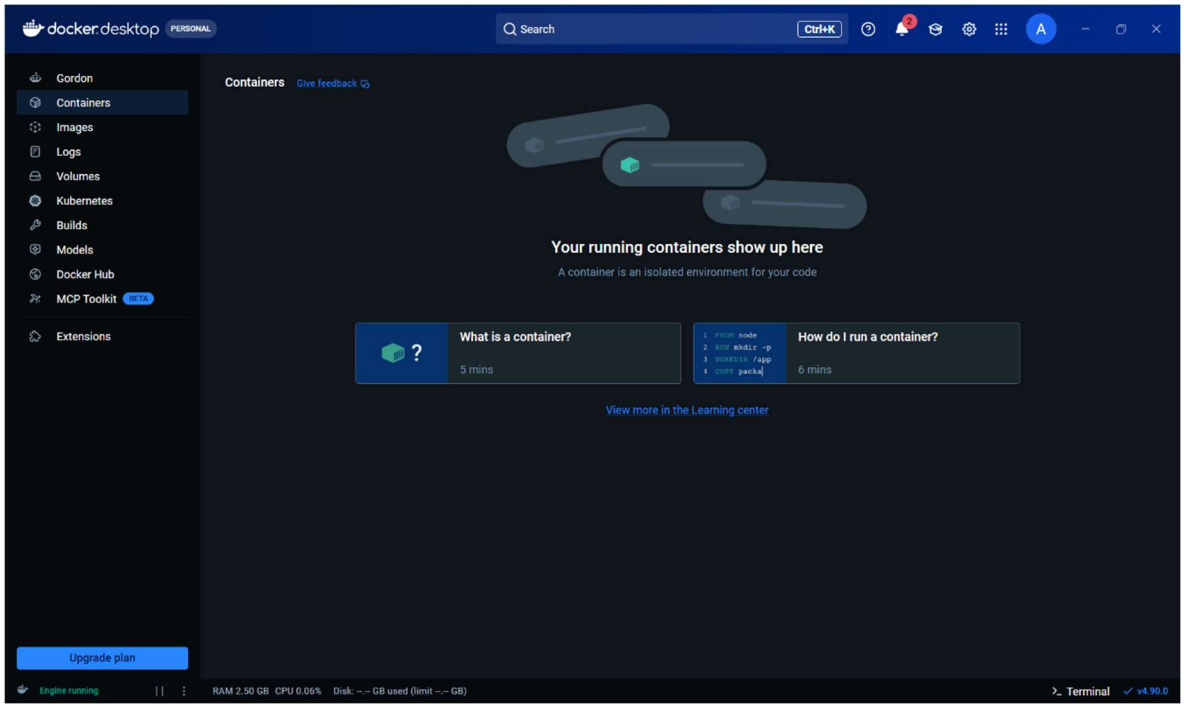
Experiment No. 7

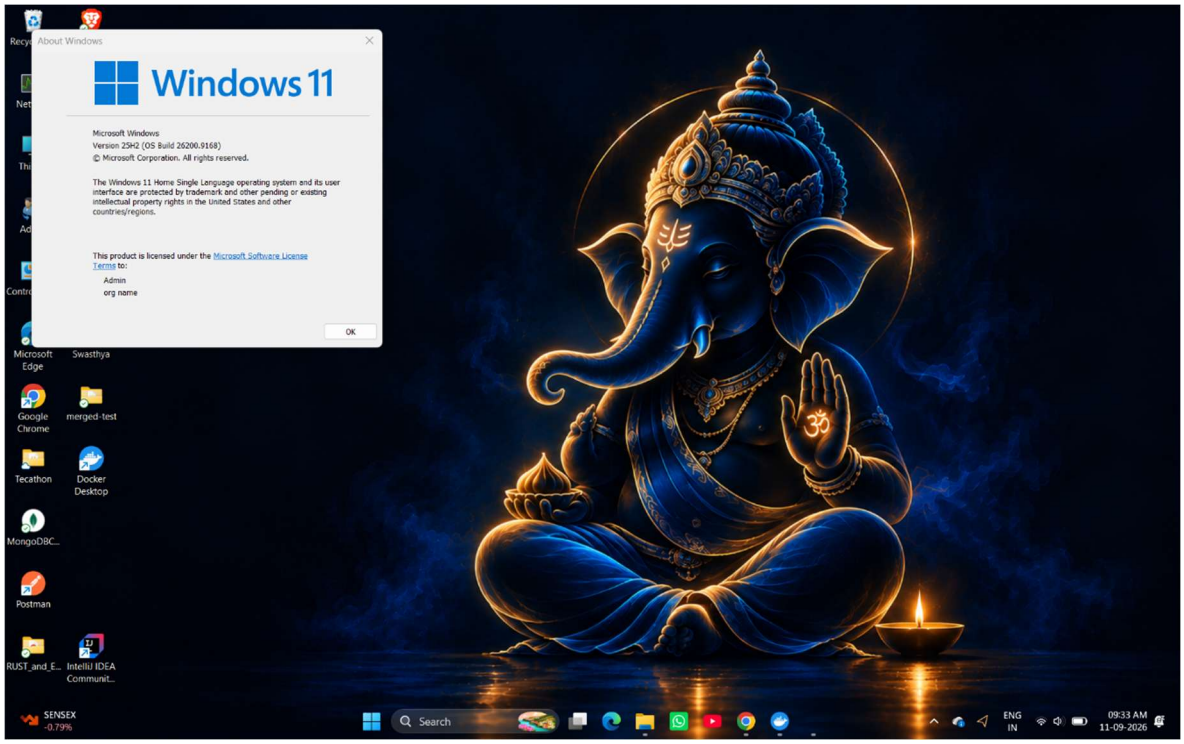
Installation and Basic Operations of Docker on Windows

NAME : Aayush Patil PRN : 2324000488

SUB : DevOps lab ROLL NO : 31 Batch: A-1

Part A: Docker installation and basic commands

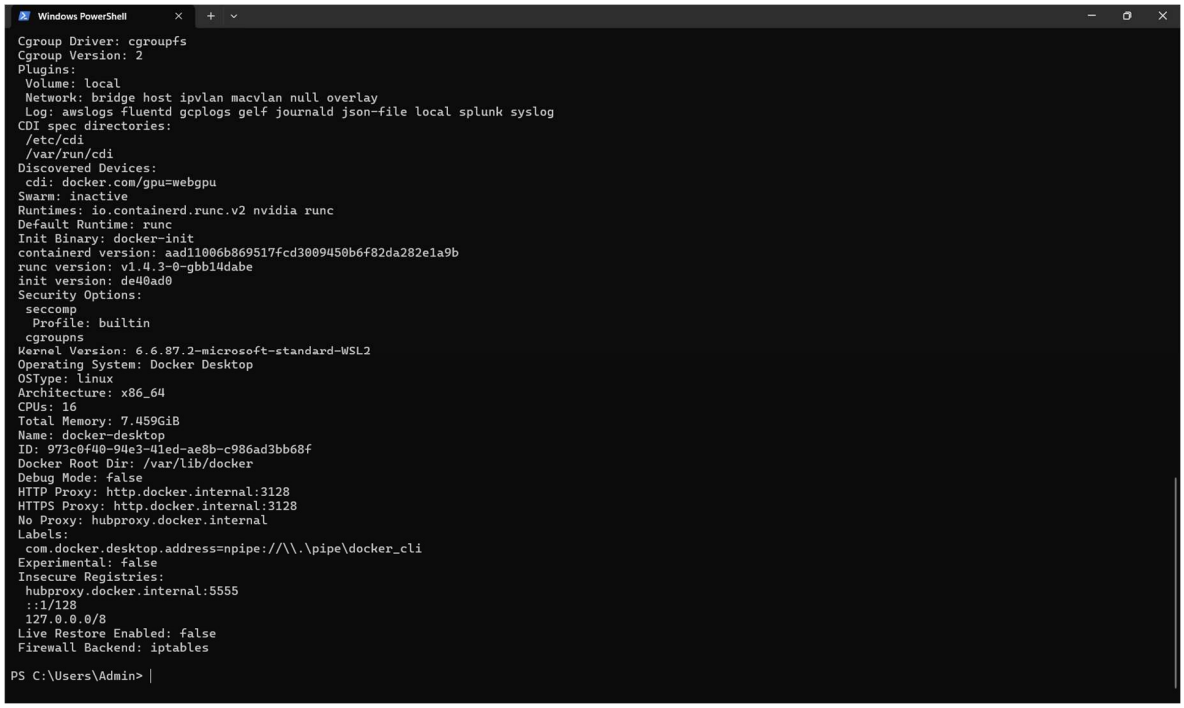


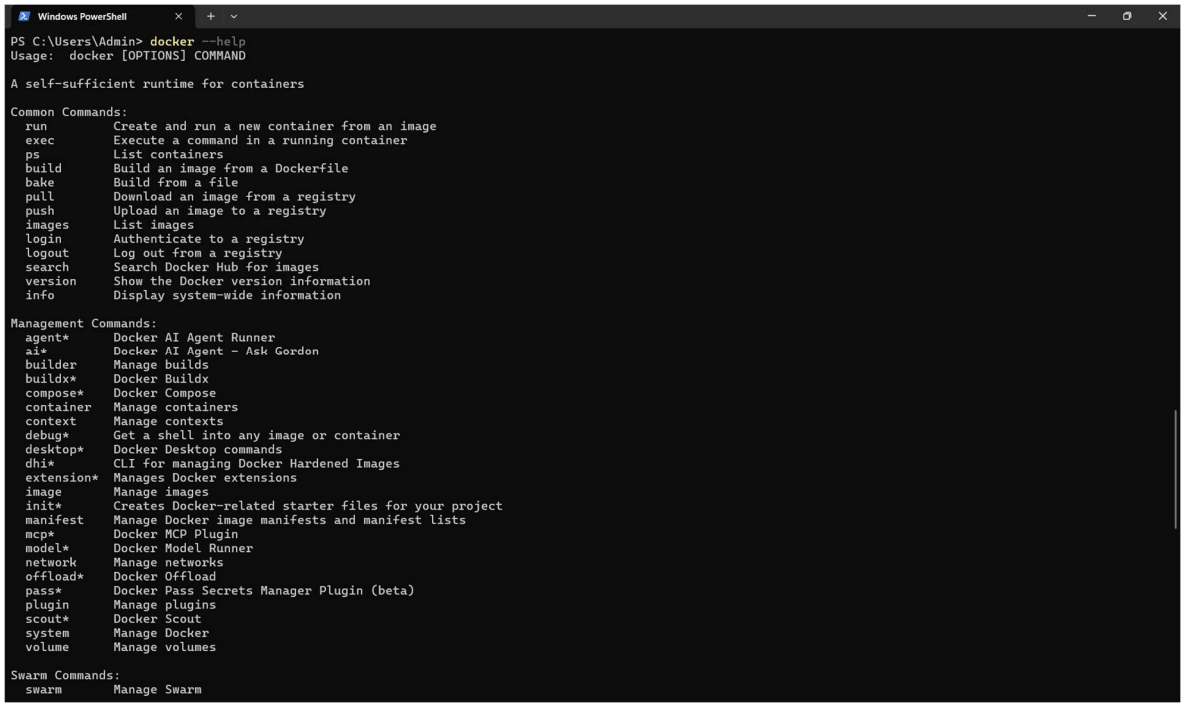



```
Windows PowerShell
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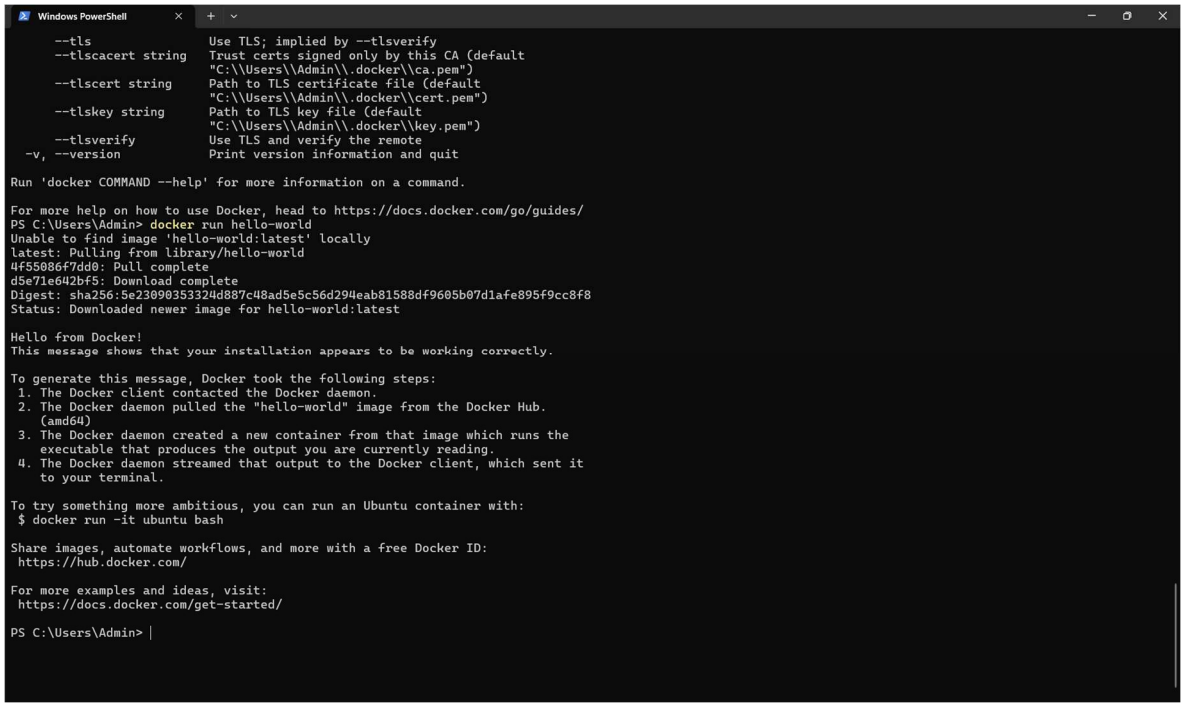
PS C:\Users\Admin> docker --version
Docker version 29.7.2, build a7dcaa6
PS C:\Users\Admin> docker version
Client:
 Version:           29.7.2
 API version:       1.55
 Go version:        go1.26.5
 Git commit:        a7dcaa6
 Built:             Wed Aug  5 18:31:33 2026
 OS/Arch:           windows/amd64
 Context:           desktop-linux

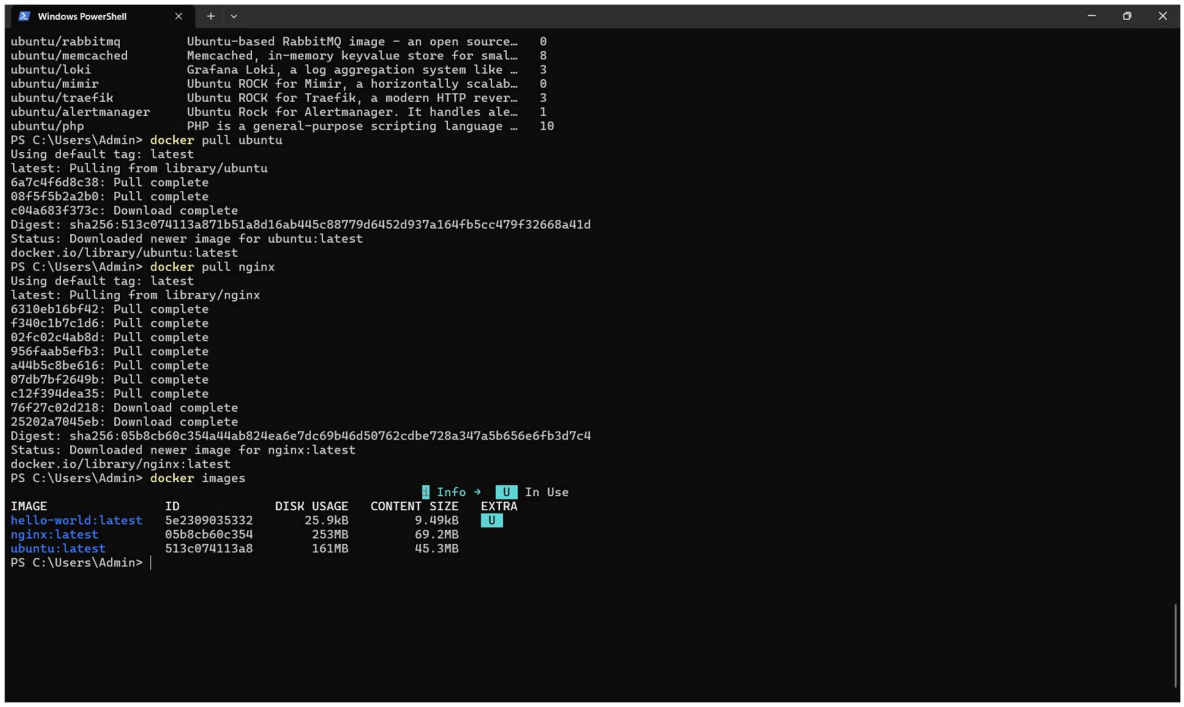
Server: Docker Desktop 4.90.0 (238679)
 Engine:
  Version:          29.7.2
  API version:      1.55 (minimum version 1.40)
  Go version:       go1.26.5
  Git commit:       6a43e3d
  Built:            Wed Aug  5 18:28:36 2026
  OS/Arch:          linux/amd64
  Experimental:     false
 containerd:
  Version:          v2.3.3
  GitCommit:        aad11006b869517fcd3009450b6f82da282e1a9b
 runc:
  Version:          1.4.3
  GitCommit:        v1.4.3-0-gbb14dabe
 docker-init:
  Version:          0.19.0
  GitCommit:        de40ad0
PS C:\Users\Admin> docker info
Client:
 Version:    29.7.2
 Context:    desktop-linux
 Debug Mode: false
 Plugins:
  agent: Docker AI Agent Runner (Docker Inc.)
    Version:  v1.128.0
    Path:     C:\Users\Admin\.docker\cli-plugins\docker-agent.exe
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version:  v1.30.0
    Path:     C:\Users\Admin\.docker\cli-plugins\docker-ai.exe
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.36.1-desktop.1
```

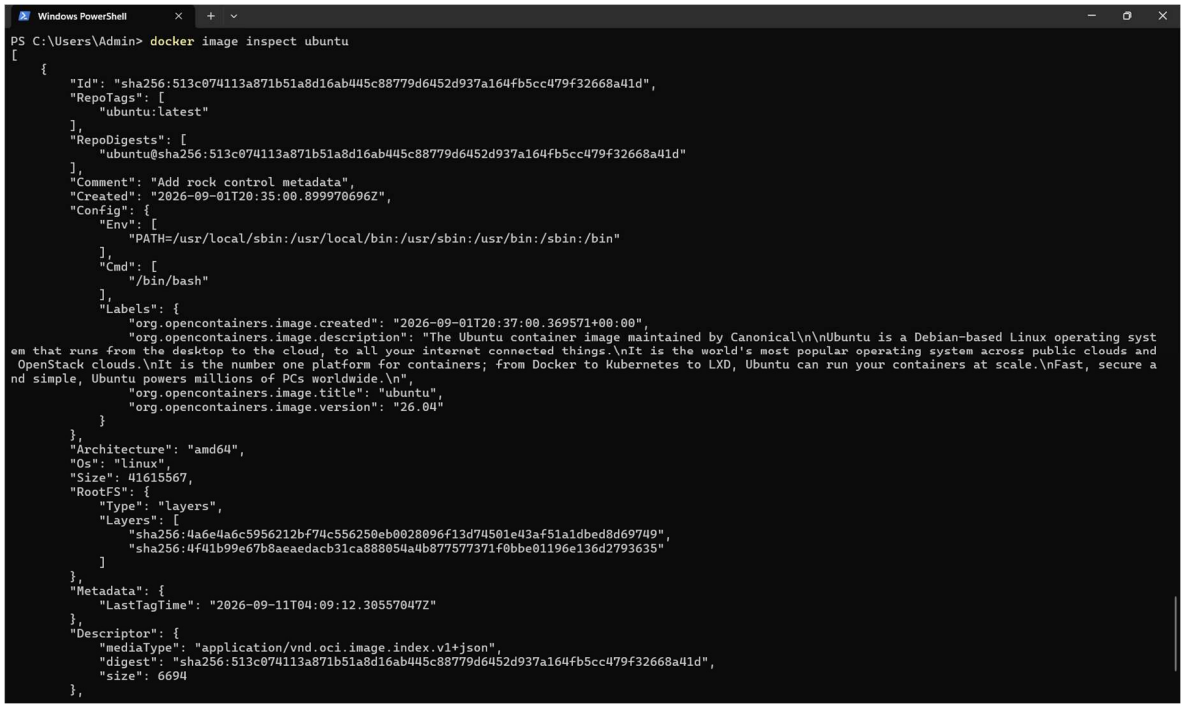

Windows PowerShell
Cgroup Driver: cgroupfs
Cgroup Version: 2
Plugins:
Volume: local
Network: bridge host ipvlan macvlan null overlay
Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
CDI spec directories:
/etc/cdi
/var/run/cdi
Discovered Devices:
cdi: docker.com/gpu=webgpu
Swarm: inactive
Runtimes: io.containerd.runc.v2 nvidia runc
Default Runtime: runc
Init Binary: docker-init
containerd version: aad11006b869517fcd3009450b6f82da282e1a9b
runc version: v1.4.3-0-gbb14dabe
init version: de40ad0
Security Options:
seccomp
Profile: builtin
cgroupns
Kernel Version: 6.6.87.2-microsoft-standard-WSL2
Operating System: Docker Desktop
OSType: linux
Architecture: x86_64
CPUs: 16
Total Memory: 7.459GiB
Name: docker-desktop
ID: 973c0f40-94e3-41ed-ae8b-c986ad3bb68f
Docker Root Dir: /var/lib/docker
Debug Mode: false
HTTP Proxy: http.docker.internal:3128
HTTPS Proxy: http.docker.internal:3128
No Proxy: hubproxy.docker.internal
Labels:
com.docker.desktop.address=npipe://\\.\pipe\docker_cli
Experimental: false
Insecure Registries:
hubproxy.docker.internal:5555
::1/128
127.0.0.0/8
Live Restore Enabled: false
Firewall Backend: iptables
PS C:\Users\Admin>

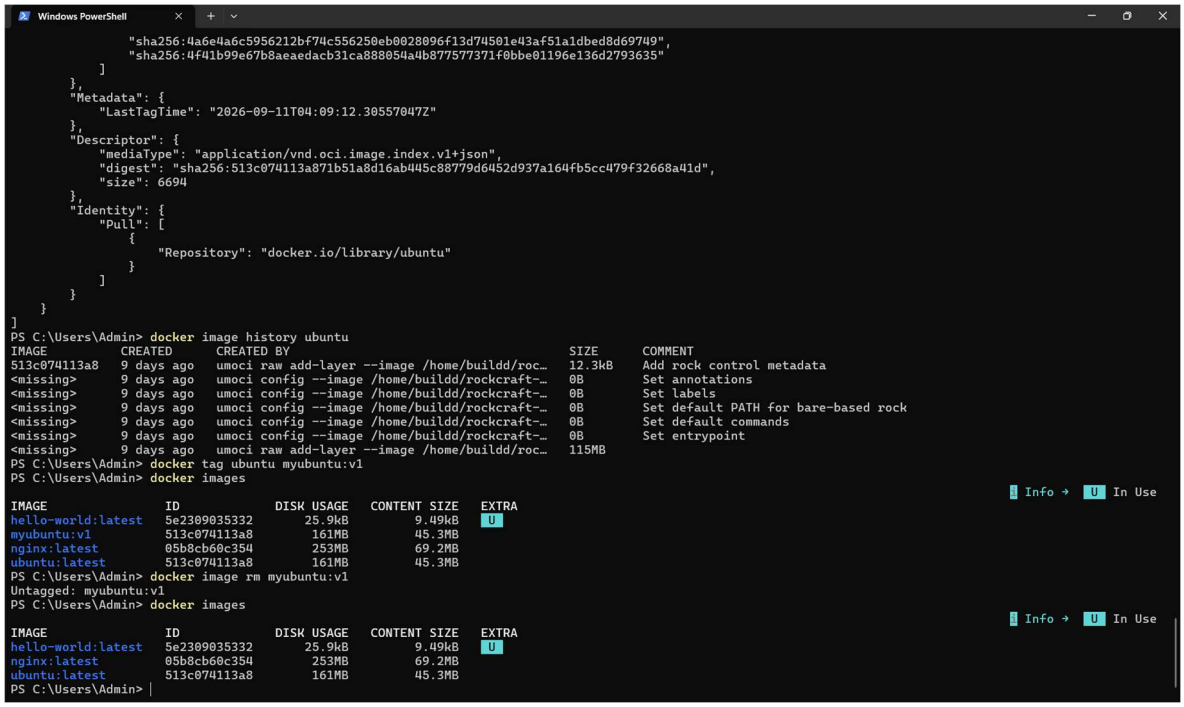

Windows PowerShell
PS C:\Users\Admin> docker --help
Usage: docker [OPTIONS] COMMAND
A self-sufficient runtime for containers
Common Commands:
run Create and run a new container from an image
exec Execute a command in a running container
ps List containers
build Build an image from a Dockerfile
bake Build from a file
pull Download an image from a registry
push Upload an image to a registry
images List images
login Authenticate to a registry
logout Log out from a registry
search Search Docker Hub for images
version Show the Docker version information
info Display system-wide information
Management Commands:
agent* Docker AI Agent Runner
ai* Docker AI Agent - Ask Gordon
builder Manage builds
buildx* Docker Buildx
compose* Docker Compose
container Manage containers
context Manage contexts
debug* Get a shell into any image or container
desktop* Docker Desktop commands
dhi* CLI for managing Docker Hardened Images
extension* Manages Docker extensions
image Manage images
init* Creates Docker-related starter files for your project
manifest Manage Docker image manifests and manifest lists
mcp* Docker MCP Plugin
model* Docker Model Runner
network Manage networks
offload* Docker Offload
pass* Docker Pass Secrets Manager Plugin (beta)
plugin Manage plugins
scout* Docker Scout
system Manage Docker
volume Manage volumes
Swarm Commands:
swarm Manage Swarm

Part B: Docker Image Commands

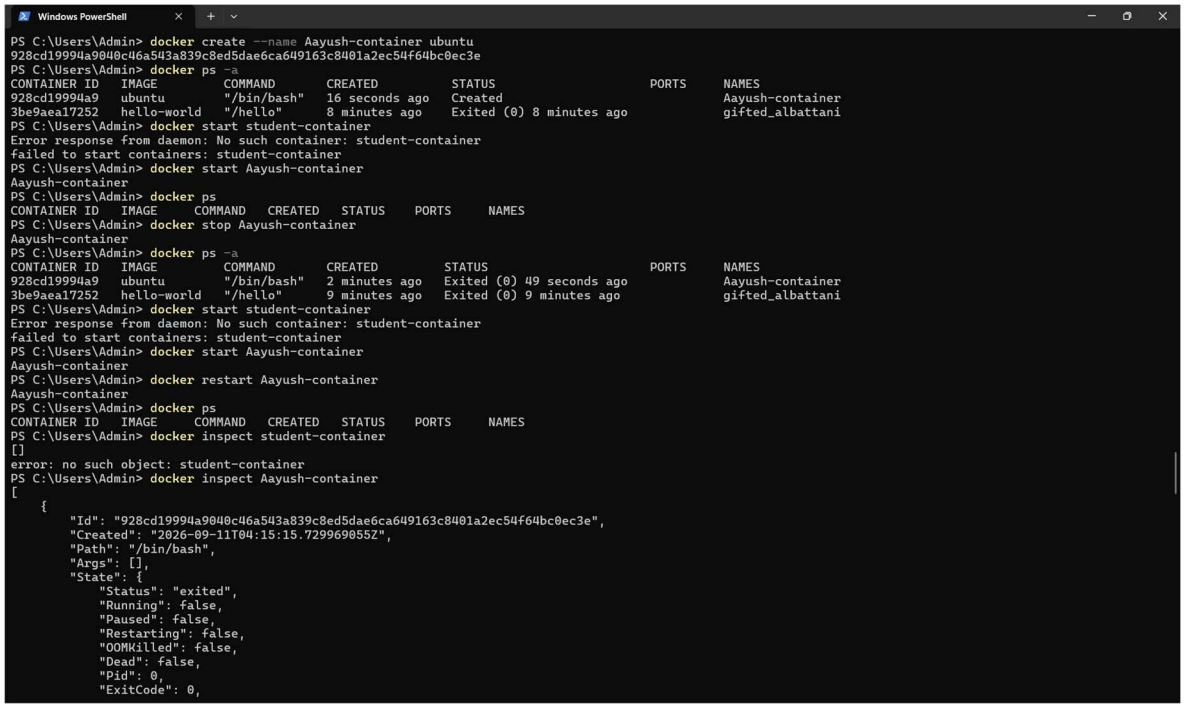




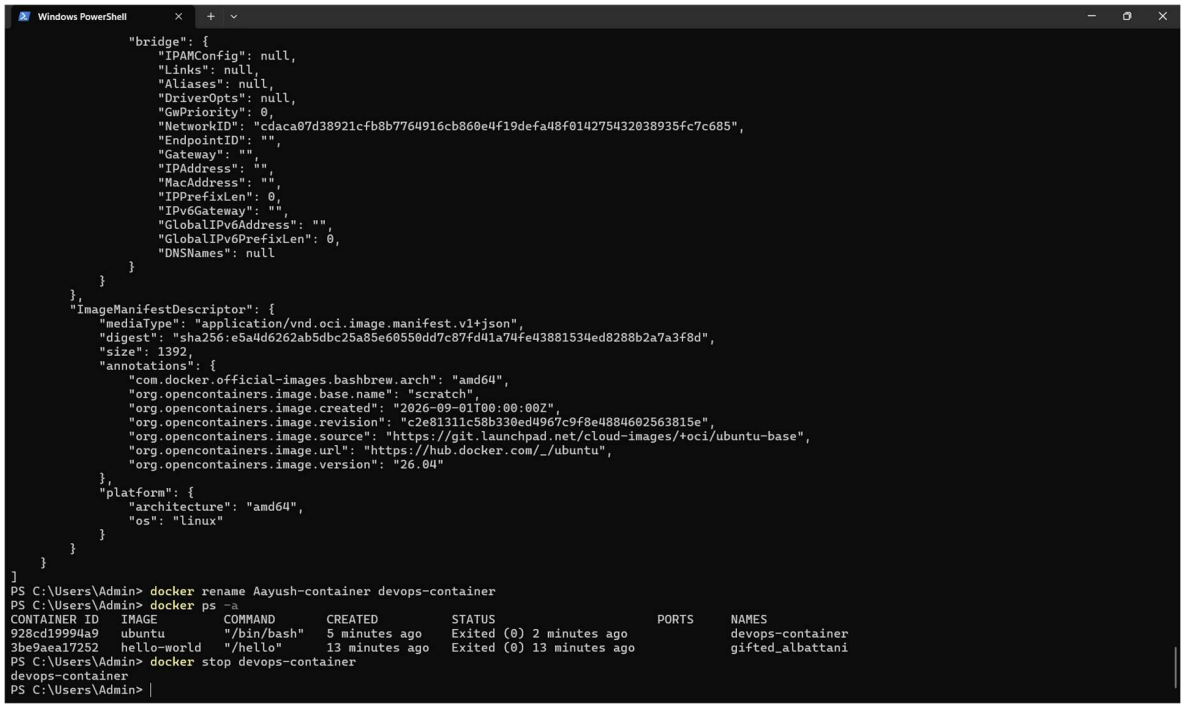

Windows PowerShell
PS C:\Users\Admin> docker image inspect ubuntu
[
{
"Id": "sha256:513c074113a871b51a8d16ab445c88779d6452d937a164fb5cc479f32668a41d",
"RepoTags": [
"ubuntu:latest"
],
"RepoDigests": [
"ubuntu@sha256:513c074113a871b51a8d16ab445c88779d6452d937a164fb5cc479f32668a41d"
],
"Comment": "Add rock control metadata",
"Created": "2026-09-01T20:35:00.899970696Z",
"Config": {
"Env": [
"PATH=/usr/local/sbin:/usr/local/bin:/usr/sbin:/usr/bin:/sbin:/bin"
],
"Cmd": [
"/bin/bash"
],
"Labels": {
"org.opencontainers.image.created": "2026-09-01T20:37:00.369571+00:00",
"org.opencontainers.image.description": "The Ubuntu container image maintained by Canonical\n\nUbuntu is a Debian-based Linux operating syst
em that runs from the desktop to the cloud, to all your internet connected things.\nIt is the world's most popular operating system across public clouds and
OpenStack clouds.\nIt is the number one platform for containers; from Docker to Kubernetes to LXD, Ubuntu can run your containers at scale.\nFast, secure a
nd simple, Ubuntu powers millions of PCs worldwide.\n",
"org.opencontainers.image.title": "ubuntu",
"org.opencontainers.image.version": "26.04"
}
},
"Architecture": "amd64",
"Os": "linux",
"Size": 41615567,
"RootFS": {
"Type": "layers",
"Layers": [
"sha256:4a6e4a6c5956212bf74c556250eb0028096f13d74501e43af51a1dbed8d69749",
"sha256:4f41b99e67b8aeaedacb31ca888054a4b877577371f0bbe01196e136d2793635"
]
},
"Metadata": {
"LastTagTime": "2026-09-11T04:09:12.305570047Z"
},
"Descriptor": {
"mediaType": "application/vnd.oci.image.index.v1+json",
"digest": "sha256:513c074113a871b51a8d16ab445c88779d6452d937a164fb5cc479f32668a41d",
"size": 6694
},


Windows PowerShell
"sha256:4a6e4a6c5956212bf74c556250eb0028096f13d74501e43af51a1dbed8d69749",
"sha256:4f41b99e67b8aeaedacb31ca888054a4b877577371f0bbe01196e136d2793635"
]
},
"Metadata": {
"LastTagTime": "2026-09-11T04:09:12.305570047Z"
},
"Descriptor": {
"mediaType": "application/vnd.oci.image.index.v1+json",
"digest": "sha256:513c074113a871b51a8d16ab445c88779d6452d937a164fb5cc479f32668a41d",
"size": 6694
},
"Identity": {
"Pull": [
{
"Repository": "docker.io/library/ubuntu"
}
]
}
}
]
PS C:\Users\Admin> docker image history ubuntu
IMAGE CREATED CREATED BY SIZE COMMENT
513c074113a8 9 days ago umoci raw add-layer --image /home/buildd/roc… 12.3kB Add rock control metadata
<missing> 9 days ago umoci config --image /home/buildd/rockcraft-… 0B Set annotations
<missing> 9 days ago umoci config --image /home/buildd/rockcraft-… 0B Set labels
<missing> 9 days ago umoci config --image /home/buildd/rockcraft-… 0B Set default PATH for bare-based rock
<missing> 9 days ago umoci config --image /home/buildd/rockcraft-… 0B Set default commands
<missing> 9 days ago umoci config --image /home/buildd/rockcraft-… 0B Set entrypoint
<missing> 9 days ago umoci raw add-layer --image /home/buildd/roc… 115MB
PS C:\Users\Admin> docker tag ubuntu myubuntu:v1
PS C:\Users\Admin> docker images
i Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
hello-world:latest 5e2309035332 25.9kB 9.49kB U
myubuntu:v1 513c074113a8 161MB 45.3MB
nginx:latest 05b8cb60c354 253MB 69.2MB
ubuntu:latest 513c074113a8 161MB 45.3MB
PS C:\Users\Admin> docker image rm myubuntu:v1
Untagged: myubuntu:v1
PS C:\Users\Admin> docker images
i Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
hello-world:latest 5e2309035332 25.9kB 9.49kB U
nginx:latest 05b8cb60c354 253MB 69.2MB
ubuntu:latest 513c074113a8 161MB 45.3MB
PS C:\Users\Admin>

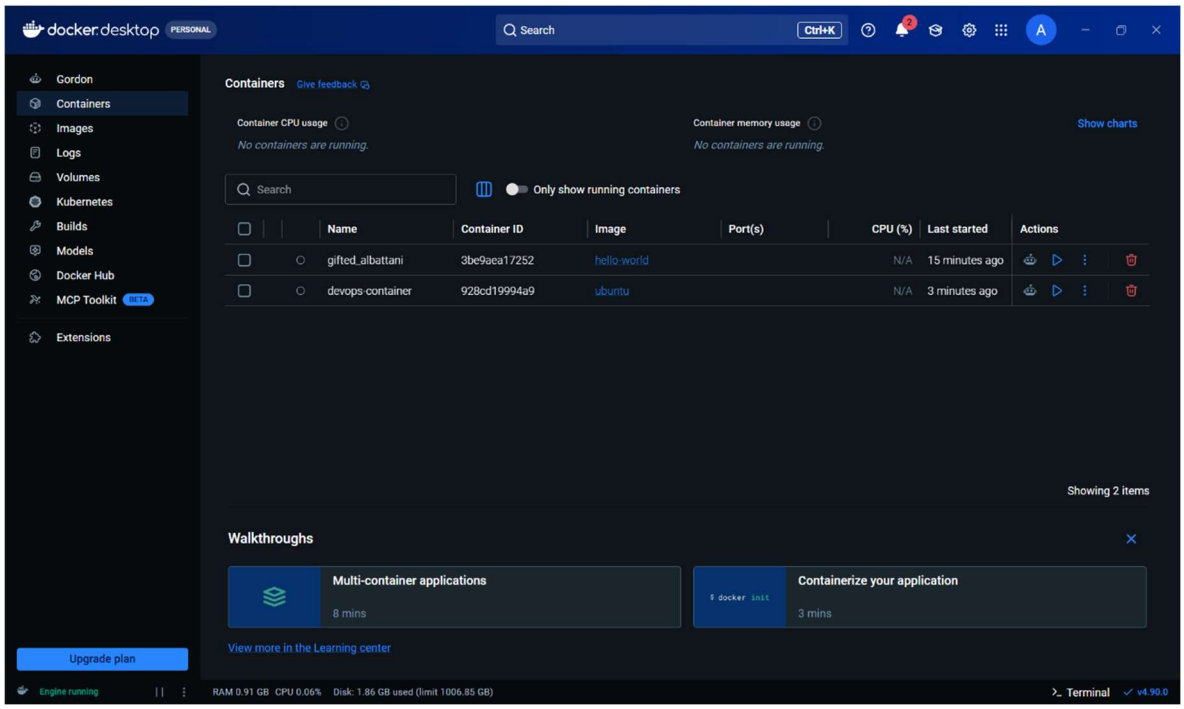
Part C: Docker Container Commands

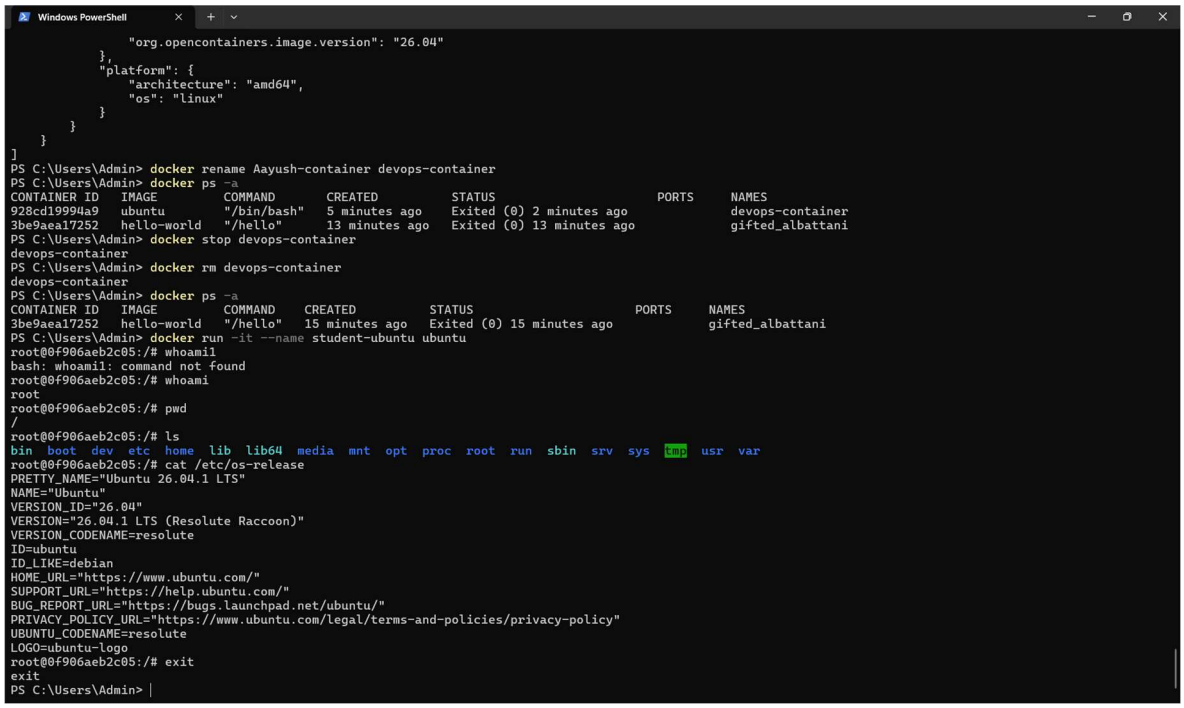


```
Windows PowerShell
PS C:\Users\Admin> docker create --name Aayush-container ubuntu
928cd19994a9040c46a543a839c8ed5dae6ca649163c8401a2ec54f64bc0ec3e
PS C:\Users\Admin> docker ps -a
CONTAINER ID   IMAGE         COMMAND       CREATED          STATUS                      PORTS     NAMES
928cd19994a9   ubuntu        "/bin/bash"   16 seconds ago   Created                               Aayush-container
3be9aea17252   hello-world   "/hello"      8 minutes ago    Exited (0) 8 minutes ago              gifted_albattani
PS C:\Users\Admin> docker start student-container
Error response from daemon: No such container: student-container
failed to start containers: student-container
PS C:\Users\Admin> docker start Aayush-container
Aayush-container
PS C:\Users\Admin> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\Admin> docker stop Aayush-container
Aayush-container
PS C:\Users\Admin> docker ps -a
CONTAINER ID   IMAGE         COMMAND       CREATED         STATUS                      PORTS     NAMES
928cd19994a9   ubuntu        "/bin/bash"   2 minutes ago   Exited (0) 49 seconds ago             Aayush-container
3be9aea17252   hello-world   "/hello"      9 minutes ago   Exited (0) 9 minutes ago              gifted_albattani
PS C:\Users\Admin> docker start student-container
Error response from daemon: No such container: student-container
failed to start containers: student-container
PS C:\Users\Admin> docker start Aayush-container
Aayush-container
PS C:\Users\Admin> docker restart Aayush-container
Aayush-container
PS C:\Users\Admin> docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
PS C:\Users\Admin> docker inspect student-container
[]
error: no such object: student-container
PS C:\Users\Admin> docker inspect Aayush-container
[
    {
        "Id": "928cd19994a9040c46a543a839c8ed5dae6ca649163c8401a2ec54f64bc0ec3e",
        "Created": "2026-09-11T04:15:15.729969055Z",
        "Path": "/bin/bash",
        "Args": [],
        "State": {
            "Status": "exited",
            "Running": false,
            "Paused": false,
            "Restarting": false,
            "OOMKilled": false,
            "Dead": false,
            "Pid": 0,
            "ExitCode": 0,
```

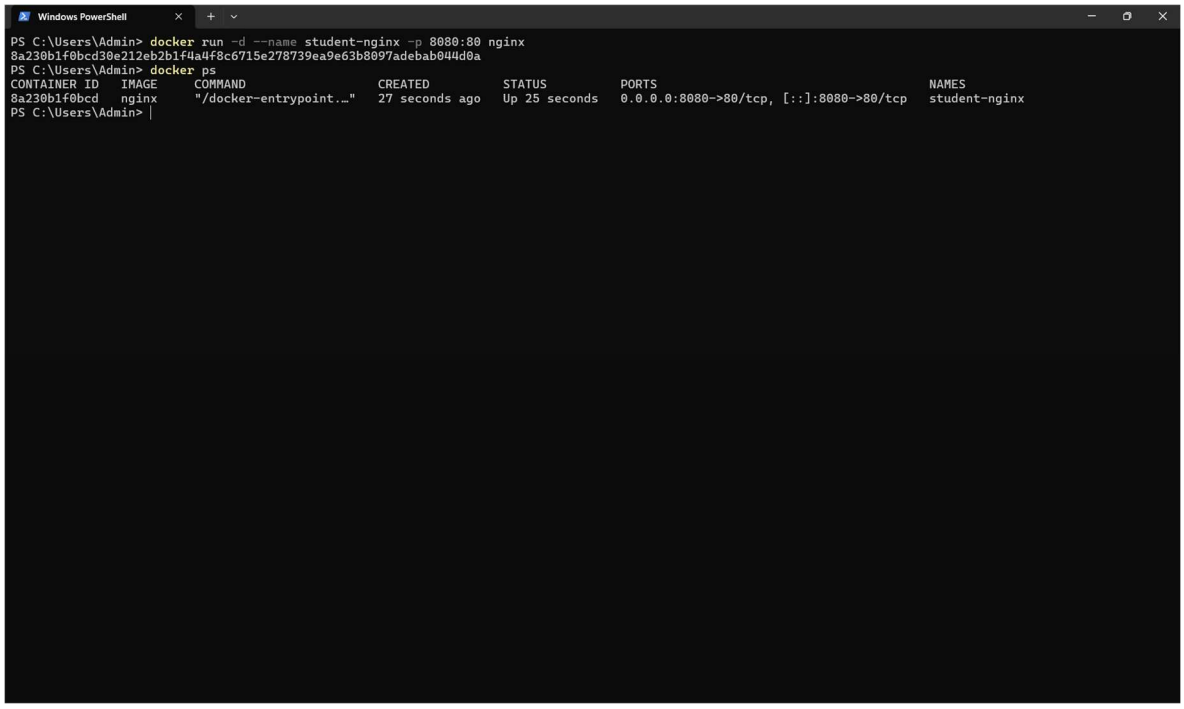


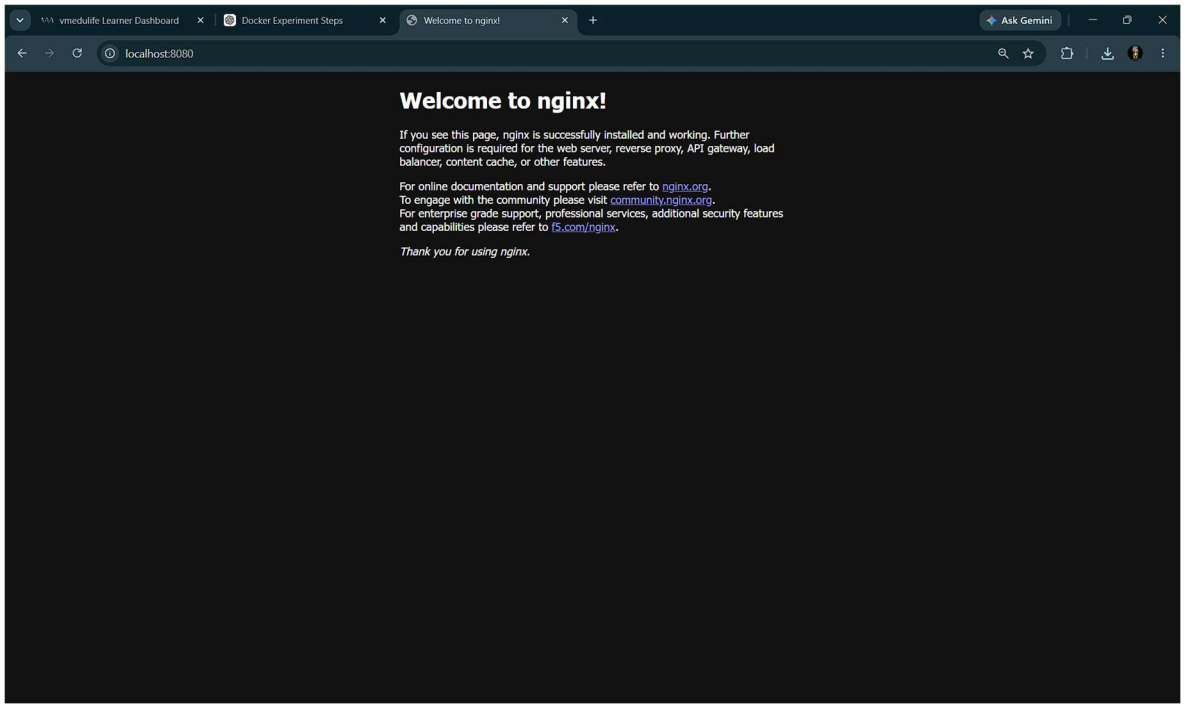
```
Windows PowerShell
                "bridge": {
                    "IPAMConfig": null,
                    "Links": null,
                    "Aliases": null,
                    "DriverOpts": null,
                    "GwPriority": 0,
                    "NetworkID": "cdaca07d38921cfb8b7764916cb860e4f19defa48f0142754320389 35fc7c685",
                    "EndpointID": "",
                    "Gateway": "",
                    "IPAddress": "",
                    "MacAddress": "",
                    "IPPrefixLen": 0,
                    "IPv6Gateway": "",
                    "GlobalIPv6Address": "",
                    "GlobalIPv6PrefixLen": 0,
                    "DNSNames": null
                }
            }
        },
        "ImageManifestDescriptor": {
            "mediaType": "application/vnd.oci.image.manifest.v1+json",
            "digest": "sha256:e5a4d6262ab5dbc25a85e60550dd7c87fd41a74fe43881534ed8288b2a7a3f8d",
            "size": 1392,
            "annotations": {
                "com.docker.official-images.bashbrew.arch": "amd64",
                "org.opencontainers.image.base.name": "scratch",
                "org.opencontainers.image.created": "2026-09-01T00:00:00Z",
                "org.opencontainers.image.revision": "c2e81311c58b330ed4967c9f8e4884602563815e",
                "org.opencontainers.image.source": "https://git.launchpad.net/cloud-images/+oci/ubuntu-base",
                "org.opencontainers.image.url": "https://hub.docker.com/_/ubuntu",
                "org.opencontainers.image.version": "26.04"
            },
            "platform": {
                "architecture": "amd64",
                "os": "linux"
            }
        }
    }
]
PS C:\Users\Admin> docker rename Aayush-container devops-container
PS C:\Users\Admin> docker ps -a
CONTAINER ID   IMAGE         COMMAND       CREATED          STATUS                      PORTS     NAMES
928cd19994a9   ubuntu        "/bin/bash"   5 minutes ago    Exited (0) 2 minutes ago              devops-container
3be9aea17252   hello-world   "/hello"      13 minutes ago   Exited (0) 13 minutes ago             gifted_albattani
PS C:\Users\Admin> docker stop devops-container
devops-container
PS C:\Users\Admin>
```

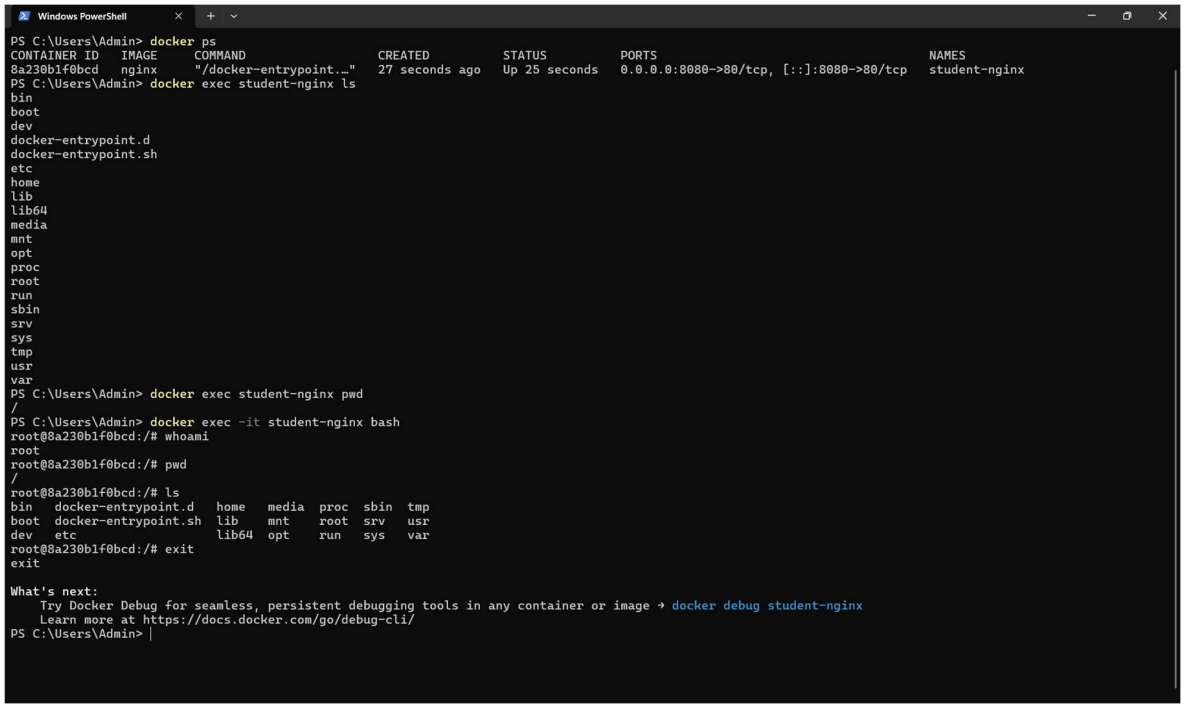

docker desktop
PERSONAL
Search
Ctrl+K
Gordon
Containers
Images
Logs
Volumes
Kubernetes
Builds
Models
Docker Hub
MCP Toolkit
BETA
Extensions
Upgrade plan
Engine running
Containers
Give feedback
Container CPU usage
No containers are running.
Container memory usage
No containers are running.
Show charts
Search
Only show running containers
Name
Container ID
Image
Port(s)
CPU (%)
Last started
Actions
gifted_albattani
3be9aea17252
hello-world
N/A
15 minutes ago
devops-container
928cd19994a9
ubuntu
N/A
3 minutes ago
Showing 2 items
Walkthroughs
Multi-container applications
8 mins
$ docker init
Containerize your application
3 mins
View more in the Learning center
RAM 0.91 GB CPU 0.06% Disk: 1.86 GB used (limit 1006.85 GB)
Terminal
v4.90.0

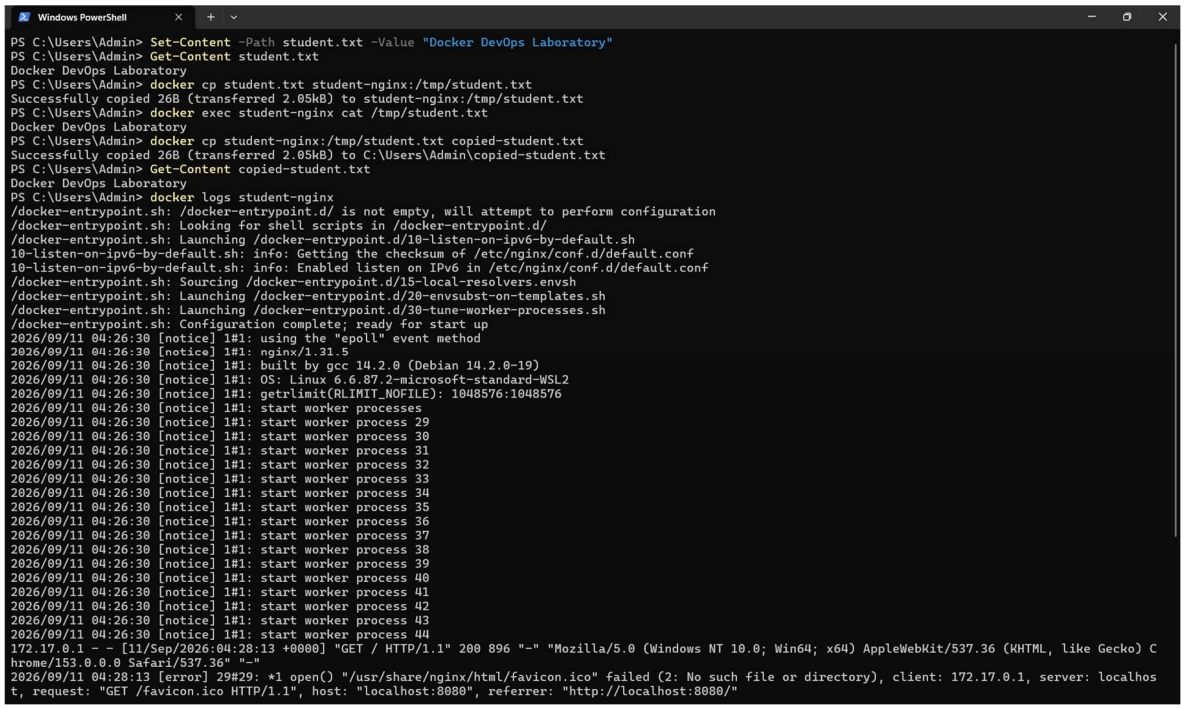

Windows PowerShell
"org.opencontainers.image.version": "26.04"
},
"platform": {
"architecture": "amd64",
"os": "linux"
}
}
}
]
PS C:\Users\Admin> docker rename Aayush-container devops-container
PS C:\Users\Admin> docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
928cd19994a9 ubuntu "/bin/bash" 5 minutes ago Exited (0) 2 minutes ago devops-container
3be9aea17252 hello-world "/hello" 13 minutes ago Exited (0) 13 minutes ago gifted_albattani
PS C:\Users\Admin> docker stop devops-container
devops-container
PS C:\Users\Admin> docker rm devops-container
devops-container
PS C:\Users\Admin> docker ps -a
CONTAINER ID IMAGE COMMAND CREATED STATUS PORTS NAMES
3be9aea17252 hello-world "/hello" 15 minutes ago Exited (0) 15 minutes ago gifted_albattani
PS C:\Users\Admin> docker run -it --name student-ubuntu ubuntu
root@0f906aeb2c05:/# whoami1
bash: whoami1: command not found
root@0f906aeb2c05:/# whoami
root
root@0f906aeb2c05:/# pwd
/
root@0f906aeb2c05:/# ls
bin boot dev etc home lib lib64 media mnt opt proc root run sbin srv sys tmp usr var
root@0f906aeb2c05:/# cat /etc/os-release
PRETTY_NAME="Ubuntu 26.04.1 LTS"
NAME="Ubuntu"
VERSION_ID="26.04"
VERSION="26.04.1 LTS (Resolute Raccoon)"
VERSION_CODENAME=resolute
ID=ubuntu
ID_LIKE=debian
HOME_URL="https://www.ubuntu.com/"
SUPPORT_URL="https://help.ubuntu.com/"
BUG_REPORT_URL="https://bugs.launchpad.net/ubuntu/"
PRIVACY_POLICY_URL="https://www.ubuntu.com/legal/terms-and-policies/privacy-policy"
UBUNTU_CODENAME=resolute
LOGO=ubuntu-logo
root@0f906aeb2c05:/# exit
exit
PS C:\Users\Admin>

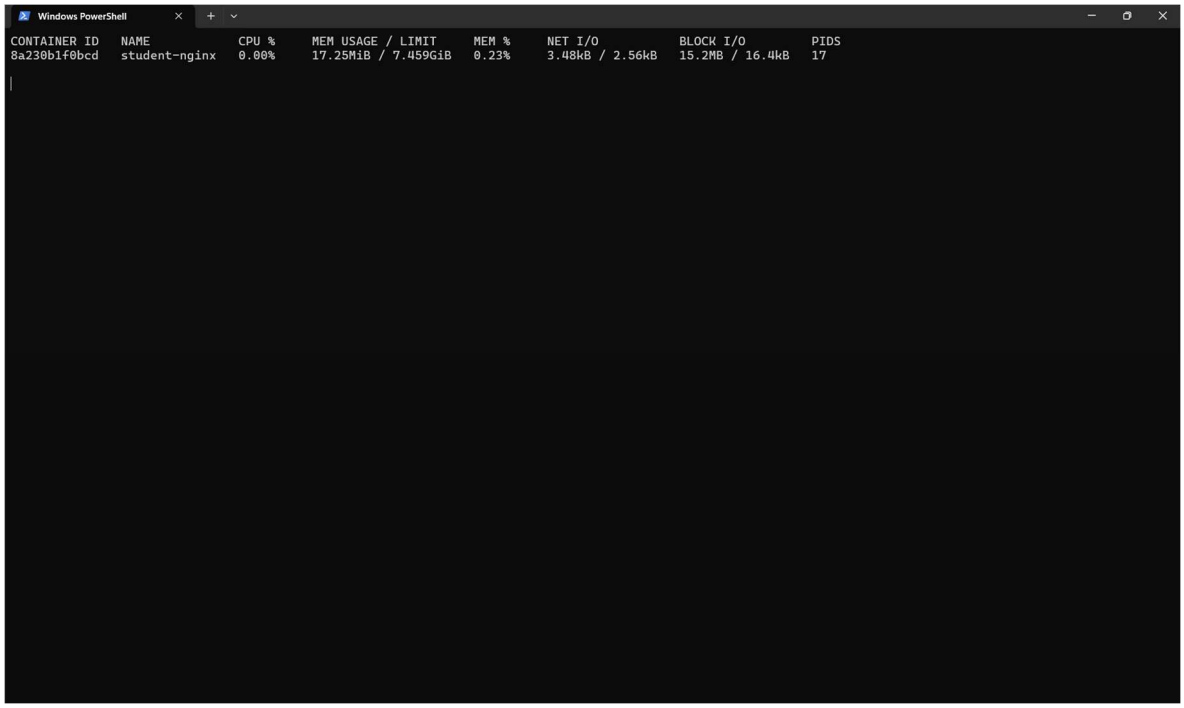
Part D: Docker Container (nginx) Commands

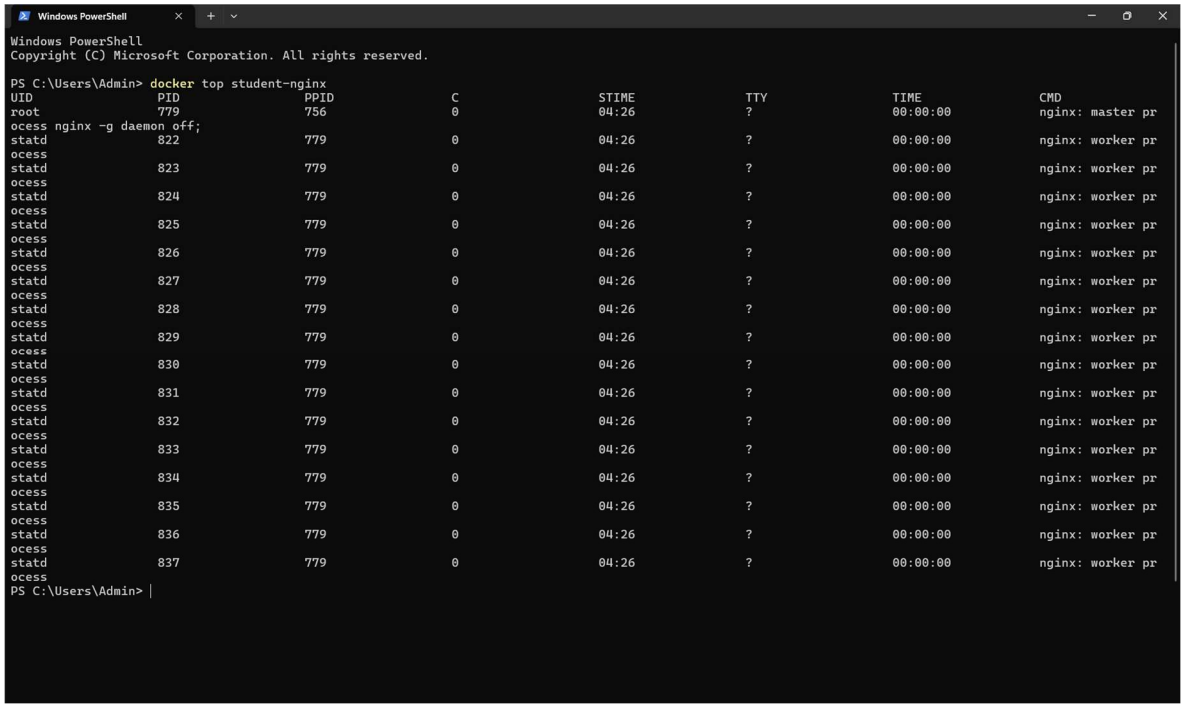


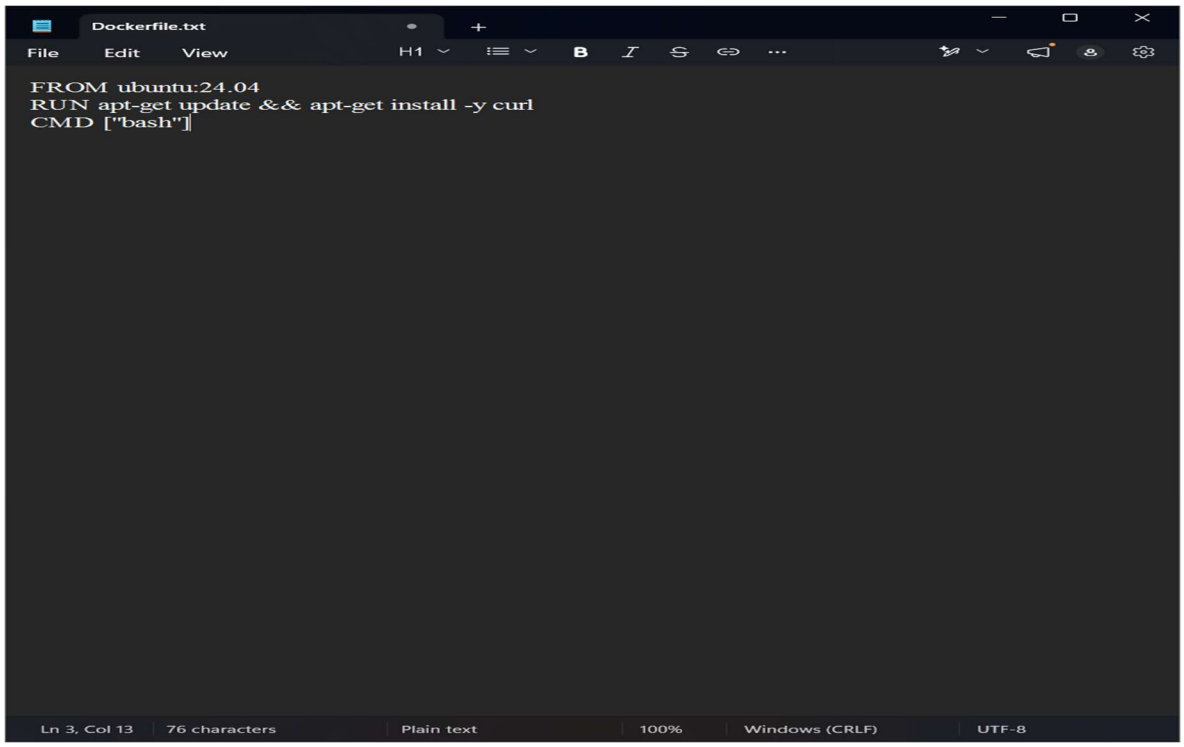


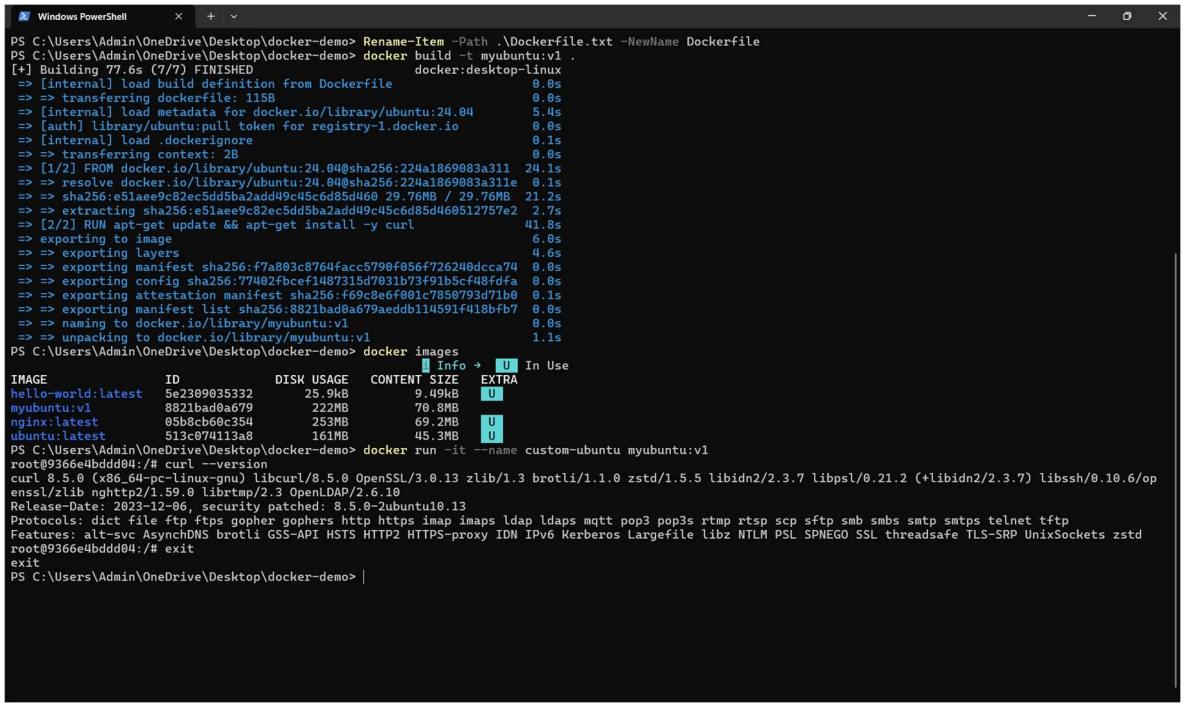

Windows PowerShell
PS C:\Users\Admin> docker ps
CONTAINER ID   IMAGE   COMMAND   CREATED   STATUS   PORTS   NAMES
8a230b1f0bcd   nginx   "/docker-entrypoint.…"   27 seconds ago   Up 25 seconds   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   student-nginx
PS C:\Users\Admin> docker exec student-nginx ls
bin
boot
dev
docker-entrypoint.d
docker-entrypoint.sh
etc
home
lib
lib64
media
mnt
opt
proc
root
run
sbin
srv
sys
tmp
usr
var
PS C:\Users\Admin> docker exec student-nginx pwd
/
PS C:\Users\Admin> docker exec -it student-nginx bash
root@8a230b1f0bcd:/# whoami
root
root@8a230b1f0bcd:/# pwd
/
root@8a230b1f0bcd:/# ls
bin   docker-entrypoint.d   home   media proc sbin tmp
boot docker-entrypoint.sh lib   mnt   root srv   usr
dev   etc   lib64 opt   run   sys   var
root@8a230b1f0bcd:/# exit
exit
What's next:
Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug student-nginx
Learn more at https://docs.docker.com/go/debug-cli/
PS C:\Users\Admin>


Windows PowerShell
PS C:\Users\Admin> Set-Content -Path student.txt -Value "Docker DevOps Laboratory"
PS C:\Users\Admin> Get-Content student.txt
Docker DevOps Laboratory
PS C:\Users\Admin> docker cp student.txt student-nginx:/tmp/student.txt
Successfully copied 26B (transferred 2.05kB) to student-nginx:/tmp/student.txt
PS C:\Users\Admin> docker exec student-nginx cat /tmp/student.txt
Docker DevOps Laboratory
PS C:\Users\Admin> docker cp student-nginx:/tmp/student.txt copied-student.txt
Successfully copied 26B (transferred 2.05kB) to C:\Users\Admin\copied-student.txt
PS C:\Users\Admin> Get-Content copied-student.txt
Docker DevOps Laboratory
PS C:\Users\Admin> docker logs student-nginx
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2026/09/11 04:26:30 [notice] 1#1: using the "epoll" event method
2026/09/11 04:26:30 [notice] 1#1: nginx/1.31.5
2026/09/11 04:26:30 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2026/09/11 04:26:30 [notice] 1#1: OS: Linux 6.6.87.2-microsoft-standard-WSL2
2026/09/11 04:26:30 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2026/09/11 04:26:30 [notice] 1#1: start worker processes
2026/09/11 04:26:30 [notice] 1#1: start worker process 29
2026/09/11 04:26:30 [notice] 1#1: start worker process 30
2026/09/11 04:26:30 [notice] 1#1: start worker process 31
2026/09/11 04:26:30 [notice] 1#1: start worker process 32
2026/09/11 04:26:30 [notice] 1#1: start worker process 33
2026/09/11 04:26:30 [notice] 1#1: start worker process 34
2026/09/11 04:26:30 [notice] 1#1: start worker process 35
2026/09/11 04:26:30 [notice] 1#1: start worker process 36
2026/09/11 04:26:30 [notice] 1#1: start worker process 37
2026/09/11 04:26:30 [notice] 1#1: start worker process 38
2026/09/11 04:26:30 [notice] 1#1: start worker process 39
2026/09/11 04:26:30 [notice] 1#1: start worker process 40
2026/09/11 04:26:30 [notice] 1#1: start worker process 41
2026/09/11 04:26:30 [notice] 1#1: start worker process 42
2026/09/11 04:26:30 [notice] 1#1: start worker process 43
2026/09/11 04:26:30 [notice] 1#1: start worker process 44
172.17.0.1 - - [11/Sep/2026:04:28:13 +0000] "GET / HTTP/1.1" 200 896 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/153.0.0.0 Safari/537.36" "-"
2026/09/11 04:28:13 [error] 29#29: *1 open() "/usr/share/nginx/html/favicon.ico" failed (2: No such file or directory), client: 172.17.0.1, server: localhost, request: "GET /favicon.ico HTTP/1.1", host: "localhost:8080", referrer: "http://localhost:8080/"

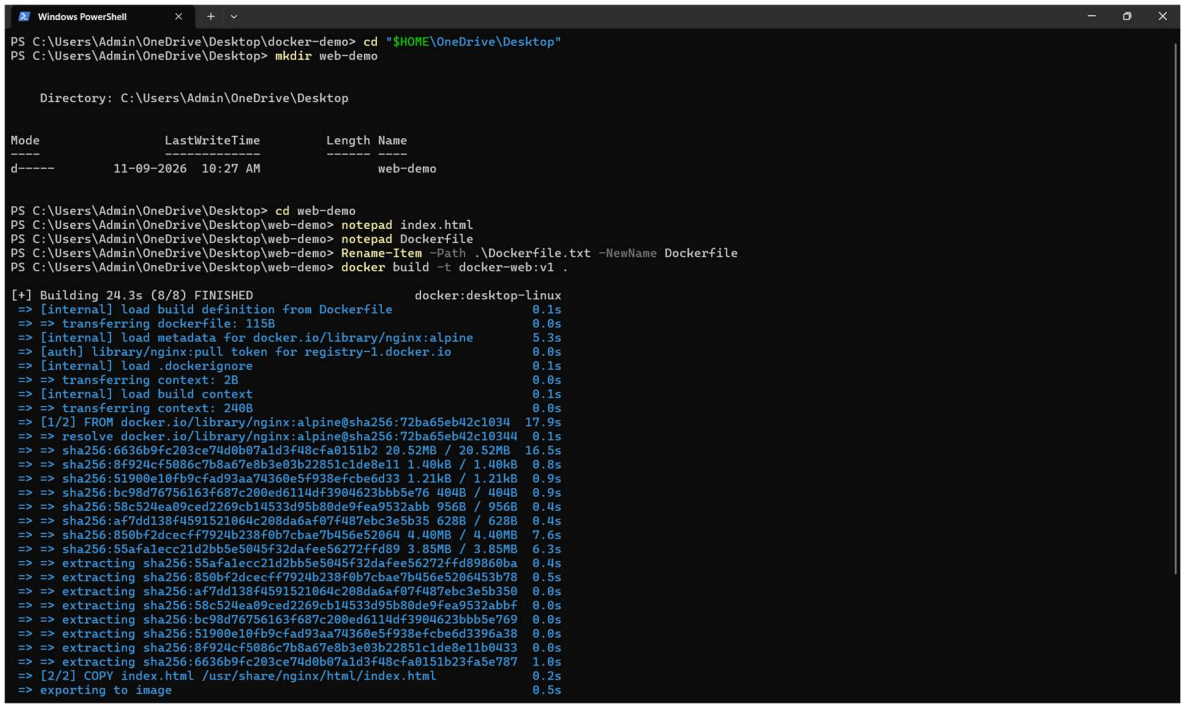

Windows PowerShell
CONTAINER ID NAME CPU % MEM USAGE / LIMIT MEM % NET I/O BLOCK I/O PIDS
8a230b1f0bcd student-nginx 0.00% 17.25MiB / 7.459GiB 0.23% 3.48kB / 2.56kB 15.2MB / 16.4kB 17


Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.
PS C:\Users\Admin> docker top student-nginx
UID PID PPID C STIME TTY TIME CMD
root 779 756 0 04:26 ? 00:00:00 nginx: master pr
ocess nginx -g daemon off;
statd 822 779 0 04:26 ? 00:00:00 nginx: worker pr
ocess
statd 823 779 0 04:26 ? 00:00:00 nginx: worker pr
ocess
statd 824 779 0 04:26 ? 00:00:00 nginx: worker pr
ocess
statd 825 779 0 04:26 ? 00:00:00 nginx: worker pr
ocess
statd 826 779 0 04:26 ? 00:00:00 nginx: worker pr
ocess
statd 827 779 0 04:26 ? 00:00:00 nginx: worker pr
ocess
statd 828 779 0 04:26 ? 00:00:00 nginx: worker pr
ocess
statd 829 779 0 04:26 ? 00:00:00 nginx: worker pr
ocess
statd 830 779 0 04:26 ? 00:00:00 nginx: worker pr
ocess
statd 831 779 0 04:26 ? 00:00:00 nginx: worker pr
ocess
statd 832 779 0 04:26 ? 00:00:00 nginx: worker pr
ocess
statd 833 779 0 04:26 ? 00:00:00 nginx: worker pr
ocess
statd 834 779 0 04:26 ? 00:00:00 nginx: worker pr
ocess
statd 835 779 0 04:26 ? 00:00:00 nginx: worker pr
ocess
statd 836 779 0 04:26 ? 00:00:00 nginx: worker pr
ocess
statd 837 779 0 04:26 ? 00:00:00 nginx: worker pr
ocess
PS C:\Users\Admin>


Dockerfile.txt
File
Edit
View
H1
FROM ubuntu:24.04
RUN apt-get update && apt-get install -y curl
CMD ["bash"]
Ln 3, Col 13
76 characters
Plain text
100%
Windows (CRLF)
UTF-8


Windows PowerShell
PS C:\Users\Admin\OneDrive\Desktop\docker-demo> Rename-Item -Path .\Dockerfile.txt -NewName Dockerfile
PS C:\Users\Admin\OneDrive\Desktop\docker-demo> docker build -t myubuntu:v1 .
[+] Building 77.6s (7/7) FINISHED docker:desktop-linux
=> [internal] load build definition from Dockerfile 0.0s
=> => transferring dockerfile: 115B 0.0s
=> [internal] load metadata for docker.io/library/ubuntu:24.04 5.4s
=> [auth] library/ubuntu:pull token for registry-1.docker.io 0.0s
=> [internal] load .dockerignore 0.1s
=> => transferring context: 2B 0.0s
=> [1/2] FROM docker.io/library/ubuntu:24.04@sha256:224a1869083a311 24.1s
=> => resolve docker.io/library/ubuntu:24.04@sha256:224a1869083a311e 0.1s
=> => sha256:e51aee9c82ec5dd5ba2add49c45c6d85d460 29.76MB / 29.76MB 21.2s
=> => extracting sha256:e51aee9c82ec5dd5ba2add49c45c6d85d460512757e2 2.7s
=> [2/2] RUN apt-get update && apt-get install -y curl 41.8s
=> exporting to image 6.0s
=> => exporting layers 4.6s
=> => exporting manifest sha256:f7a803c8764facc5790f056f726240dcca74 0.0s
=> => exporting config sha256:77402fbcef1487315d7031b73f91b5cf48fdfa 0.0s
=> => exporting attestation manifest sha256:f69c8e6f001c7850793d71b0 0.1s
=> => exporting manifest list sha256:8821bad0a679aeddb114591f418bfb7 0.0s
=> => naming to docker.io/library/myubuntu:v1 0.0s
=> => unpacking to docker.io/library/myubuntu:v1 1.1s
PS C:\Users\Admin\OneDrive\Desktop\docker-demo> docker images
Info → U In Use
IMAGE ID DISK USAGE CONTENT SIZE EXTRA
hello-world:latest 5e2309035332 25.9kB 9.49kB U
myubuntu:v1 8821bad0a679 222MB 70.8MB
nginx:latest 05b8cb60c354 253MB 69.2MB U
ubuntu:latest 513c074113a8 161MB 45.3MB U
PS C:\Users\Admin\OneDrive\Desktop\docker-demo> docker run -it --name custom-ubuntu myubuntu:v1
root@9366e4bddd04:/# curl --version
curl 8.5.0 (x86_64-pc-linux-gnu) libcurl/8.5.0 OpenSSL/3.0.13 zlib/1.3 brotli/1.1.0 zstd/1.5.5 libidn2/2.3.7 libpsl/0.21.2 (+libidn2/2.3.7) libssh/0.10.6/op
enssl/zlib nghttp2/1.59.0 librtmp/2.3 OpenLDAP/2.6.10
Release-Date: 2023-12-06, security patched: 8.5.0-2ubuntu10.13
Protocols: dict file ftp ftps gopher gophers http https imap imaps ldap ldaps mqtt pop3 pop3s rtmp rtsp scp sftp smb smbs smtp smtps telnet tftp
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NTLM PSL SPNEGO SSL threadsafe TLS-SRP UnixSockets zstd
root@9366e4bddd04:/# exit
exit
PS C:\Users\Admin\OneDrive\Desktop\docker-demo>

Part E: Docker Container (web-demo) Commands



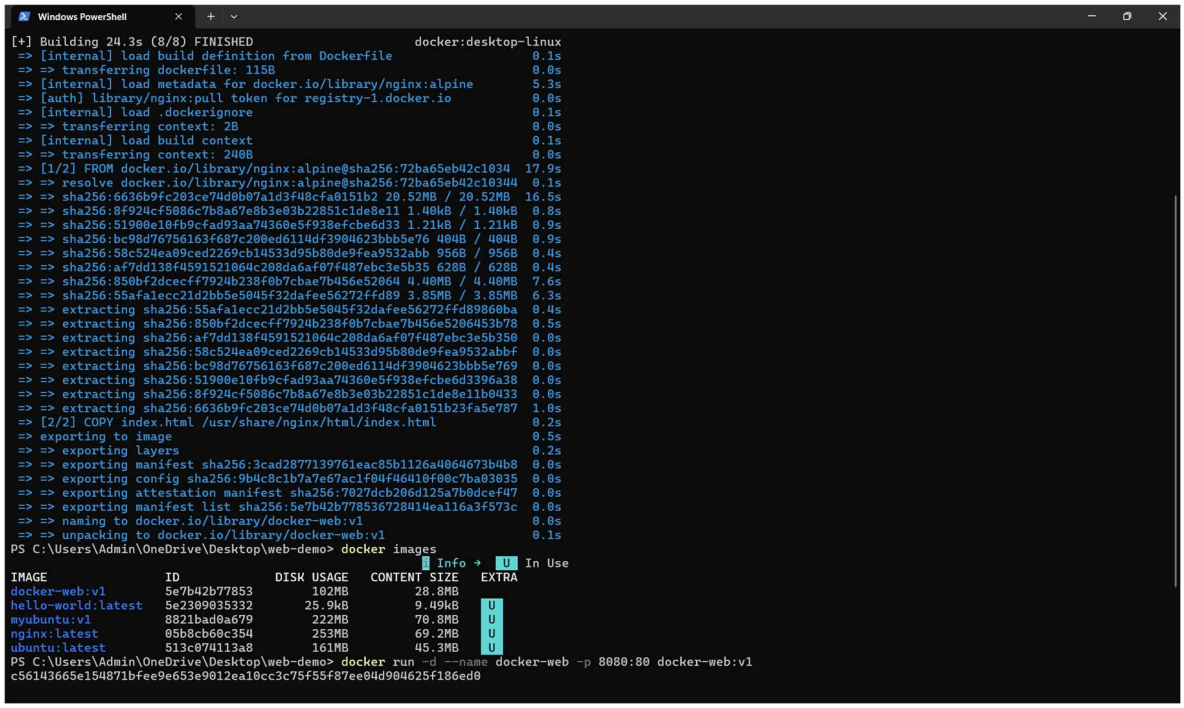
```
Windows PowerShell
PS C:\Users\Admin\OneDrive\Desktop\docker-demo> cd "$HOME\OneDrive\Desktop"
PS C:\Users\Admin\OneDrive\Desktop> mkdir web-demo


    Directory: C:\Users\Admin\OneDrive\Desktop

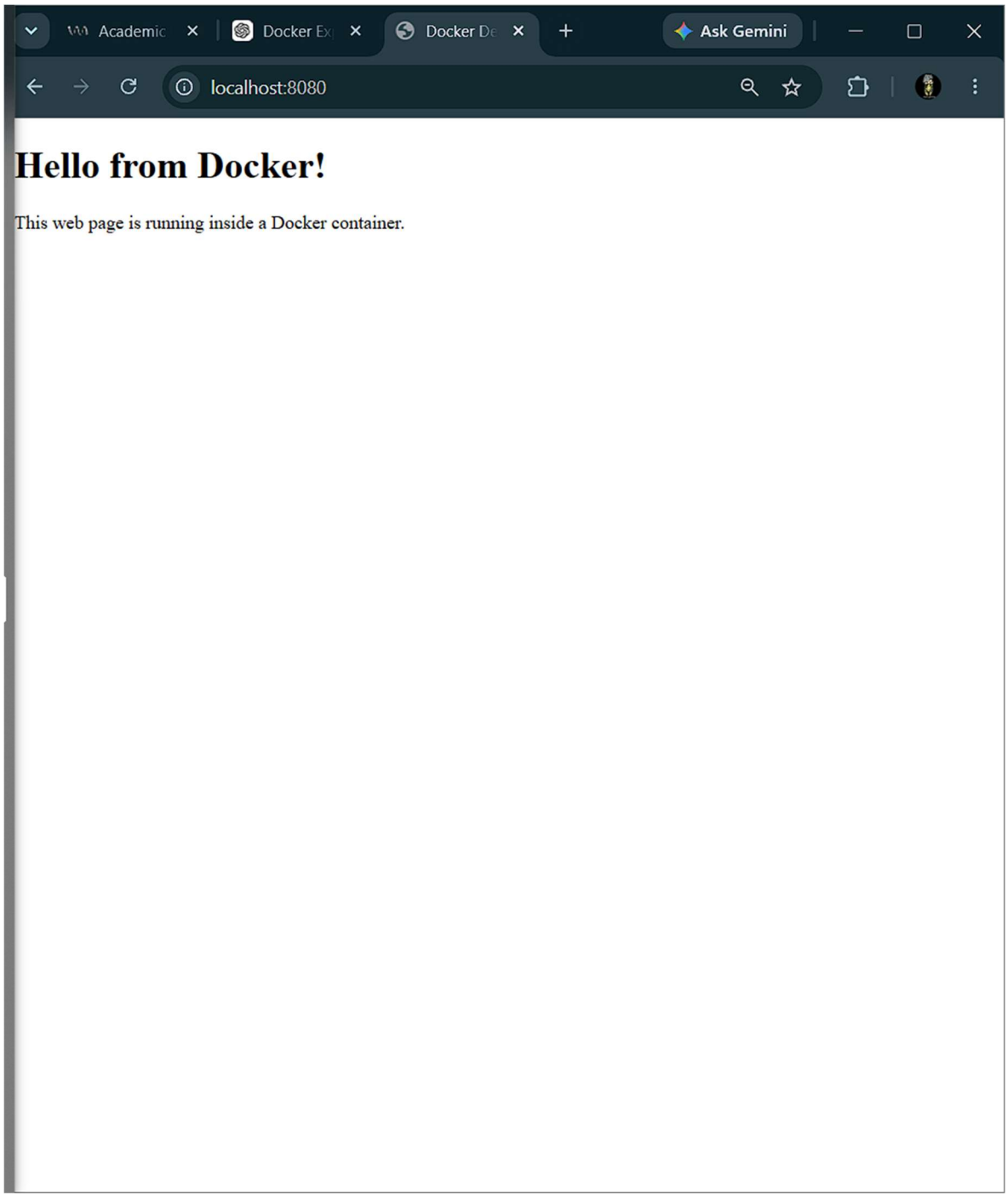

Mode                 LastWriteTime         Length Name
----                 -------------         ------ ----
d-----        11-09-2026  10:27 AM                web-demo


PS C:\Users\Admin\OneDrive\Desktop> cd web-demo
PS C:\Users\Admin\OneDrive\Desktop\web-demo> notepad index.html
PS C:\Users\Admin\OneDrive\Desktop\web-demo> notepad Dockerfile
PS C:\Users\Admin\OneDrive\Desktop\web-demo> Rename-Item -Path .\Dockerfile.txt -NewName Dockerfile
PS C:\Users\Admin\OneDrive\Desktop\web-demo> docker build -t docker-web:v1 .

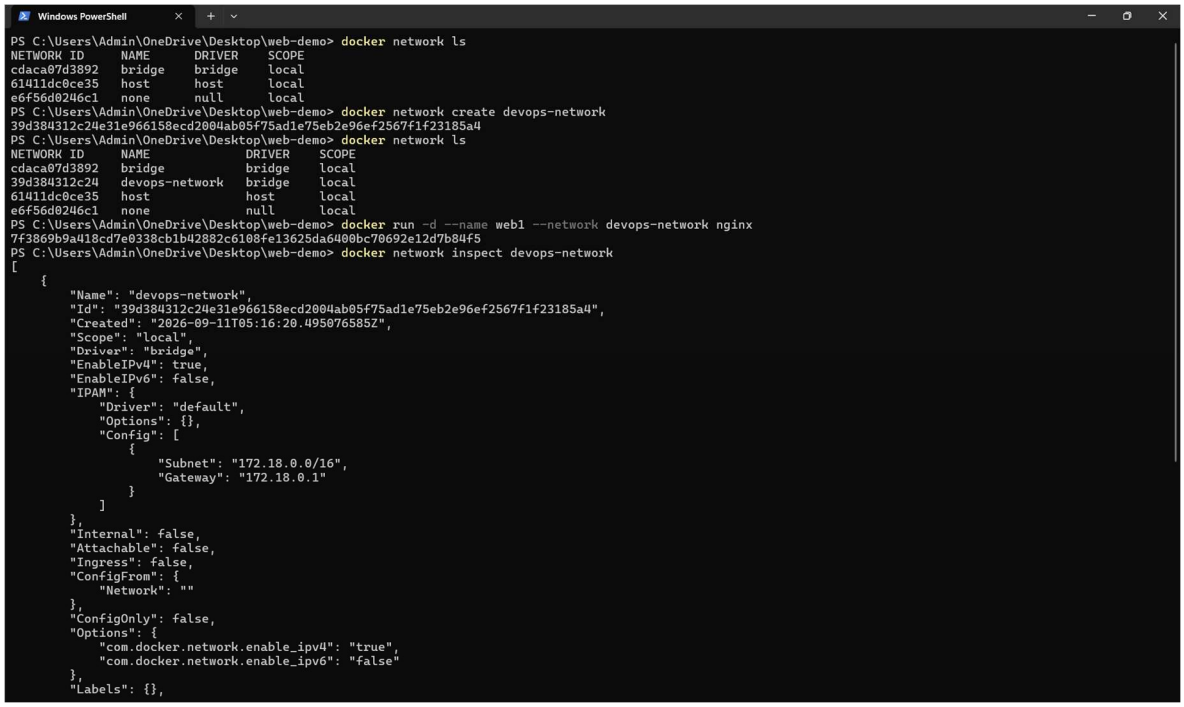
[+] Building 24.3s (8/8) FINISHED                       docker:desktop-linux
 => [internal] load build definition from Dockerfile                    0.1s
 => => transferring dockerfile: 115B                                    0.0s
 => [internal] load metadata for docker.io/library/nginx:alpine         5.3s
 => [auth] library/nginx:pull token for registry-1.docker.io            0.0s
 => [internal] load .dockerignore                                       0.1s
 => => transferring context: 2B                                         0.0s
 => [internal] load build context                                       0.1s
 => => transferring context: 240B                                       0.0s
 => [1/2] FROM docker.io/library/nginx:alpine@sha256:72ba65eb42c1034   17.9s
 => => resolve docker.io/library/nginx:alpine@sha256:72ba65eb42c10344   0.1s
 => => sha256:6636b9fc203ce74d0b07a1d3f48cfa0151b2 20.52MB / 20.52MB   16.5s
 => => sha256:8f924cf5086c7b8a67e8b3e03b22851c1de8e11 1.40kB / 1.40kB   0.8s
 => => sha256:51900e10fb9cfad93aa74360e5f938efcbe6d33 1.21kB / 1.21kB   0.9s
 => => sha256:bc98d76756163f687c200ed6114df3904623bbb5e76 404B / 404B   0.9s
 => => sha256:58c524ea09ced2269cb14533d95b80de9fea9532abb 956B / 956B   0.4s
 => => sha256:af7dd138f4591521064c208da6af07f487ebc3e5b35 628B / 628B   0.4s
 => => sha256:850bf2dcecff7924b238f0b7cbae7b456e52064 4.40MB / 4.40MB   7.6s
 => => sha256:55afa1ecc21d2bb5e5045f32dafee56272ffd89 3.85MB / 3.85MB   6.3s
 => => extracting sha256:55afa1ecc21d2bb5e5045f32dafee56272ffd89860ba   0.4s
 => => extracting sha256:850bf2dcecff7924b238f0b7cbae7b456e5206453b78   0.5s
 => => extracting sha256:af7dd138f4591521064c208da6af07f487ebc3e5b350   0.0s
 => => extracting sha256:58c524ea09ced2269cb14533d95b80de9fea9532abbf   0.0s
 => => extracting sha256:bc98d76756163f687c200ed6114df3904623bbb5e769   0.0s
 => => extracting sha256:51900e10fb9cfad93aa74360e5f938efcbe6d3396a38   0.0s
 => => extracting sha256:8f924cf5086c7b8a67e8b3e03b22851c1de8e11b0433   0.0s
 => => extracting sha256:6636b9fc203ce74d0b07a1d3f48cfa0151b23fa5e787   1.0s
 => [2/2] COPY index.html /usr/share/nginx/html/index.html              0.2s
 => exporting to image                                                  0.5s
```



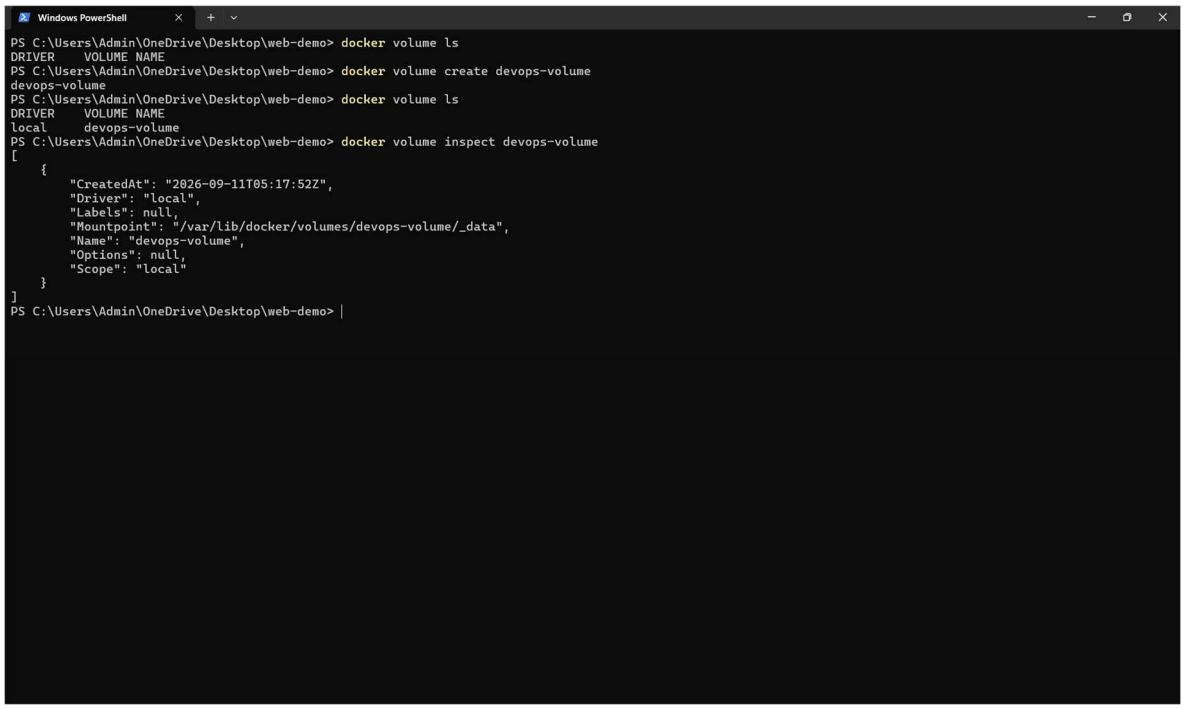
```
Windows PowerShell
[+] Building 24.3s (8/8) FINISHED                       docker:desktop-linux
 => [internal] load build definition from Dockerfile                    0.1s
 => => transferring dockerfile: 115B                                    0.0s
 => [internal] load metadata for docker.io/library/nginx:alpine         5.3s
 => [auth] library/nginx:pull token for registry-1.docker.io            0.0s
 => [internal] load .dockerignore                                       0.1s
 => => transferring context: 2B                                         0.0s
 => [internal] load build context                                       0.1s
 => => transferring context: 240B                                       0.0s
 => [1/2] FROM docker.io/library/nginx:alpine@sha256:72ba65eb42c1034   17.9s
 => => resolve docker.io/library/nginx:alpine@sha256:72ba65eb42c10344   0.1s
 => => sha256:6636b9fc203ce74d0b07a1d3f48cfa0151b2 20.52MB / 20.52MB   16.5s
 => => sha256:8f924cf5086c7b8a67e8b3e03b22851c1de8e11 1.40kB / 1.40kB   0.8s
 => => sha256:51900e10fb9cfad93aa74360e5f938efcbe6d33 1.21kB / 1.21kB   0.9s
 => => sha256:bc98d76756163f687c200ed6114df3904623bbb5e76 404B / 404B   0.9s
 => => sha256:58c524ea09ced2269cb14533d95b80de9fea9532abb 956B / 956B   0.4s
 => => sha256:af7dd138f4591521064c208da6af07f487ebc3e5b35 628B / 628B   0.4s
 => => sha256:850bf2dcecff7924b238f0b7cbae7b456e52064 4.40MB / 4.40MB   7.6s
 => => sha256:55afa1ecc21d2bb5e5045f32dafee56272ffd89 3.85MB / 3.85MB   6.3s
 => => extracting sha256:55afa1ecc21d2bb5e5045f32dafee56272ffd89860ba   0.4s
 => => extracting sha256:850bf2dcecff7924b238f0b7cbae7b456e5206453b78   0.5s
 => => extracting sha256:af7dd138f4591521064c208da6af07f487ebc3e5b350   0.0s
 => => extracting sha256:58c524ea09ced2269cb14533d95b80de9fea9532abbf   0.0s
 => => extracting sha256:bc98d76756163f687c200ed6114df3904623bbb5e769   0.0s
 => => extracting sha256:51900e10fb9cfad93aa74360e5f938efcbe6d3396a38   0.0s
 => => extracting sha256:8f924cf5086c7b8a67e8b3e03b22851c1de8e11b0433   0.0s
 => => extracting sha256:6636b9fc203ce74d0b07a1d3f48cfa0151b23fa5e787   1.0s
 => [2/2] COPY index.html /usr/share/nginx/html/index.html              0.2s
 => exporting to image                                                  0.5s
 => => exporting layers                                                 0.2s
 => => exporting manifest sha256:3cad2877139761eac85b1126a4064673b4b8   0.0s
 => => exporting config sha256:9b4c8c1b7a7e67ac1f04f46410f00c7ba03035   0.0s
 => => exporting attestation manifest sha256:7027dcb206d125a7b0dcef47   0.0s
 => => exporting manifest list sha256:5e7b42b7785367284l4ea116a3f573c   0.0s
 => => naming to docker.io/library/docker-web:v1                        0.0s
 => => unpacking to docker.io/library/docker-web:v1                     0.1s
PS C:\Users\Admin\OneDrive\Desktop\web-demo> docker images
                                                  i Info →   U  In Use
IMAGE                ID             DISK USAGE   CONTENT SIZE   EXTRA
docker-web:v1        5e7b42b77853       102MB         28.8MB
hello-world:latest   5e2309035332      25.9kB         9.49kB    U
myubuntu:v1          8821bad0a679       222MB         70.8MB    U
nginx:latest         05b8cb60c354       253MB         69.2MB    U
ubuntu:latest        513c074113a8       161MB         45.3MB    U
PS C:\Users\Admin\OneDrive\Desktop\web-demo> docker run -d --name docker-web -p 8080:80 docker-web:v1
c56143665e154871bfee9e653e9012ea10cc3c75f55f87ee04d904625f186ed0
```

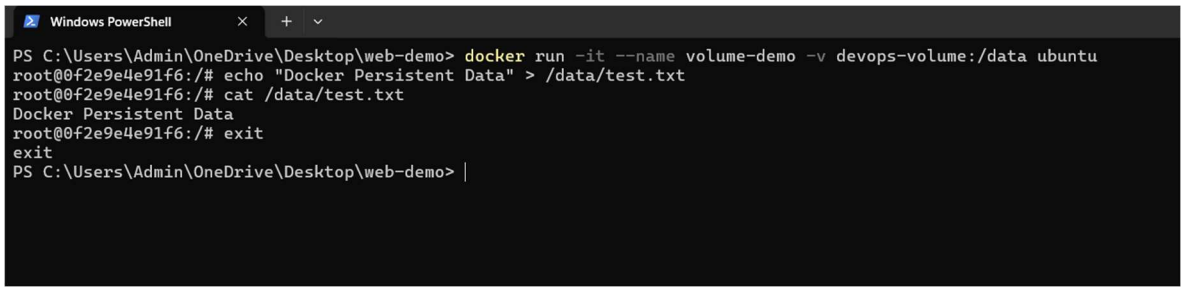

Academic
Docker Ex
Docker De
Ask Gemini
localhost:8080
Hello from Docker!
This web page is running inside a Docker container.

Part F: Docker Network Commands



Part G: Docker Volume Commands





Part H: Docker Remove/Delete Commands

